26 April 1988
SUPERSEDING
MIL-B-49430/9B(ER)
15 June 1987

MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5513/U

This specification is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the battery described herein shall consist of this specification and the latest issue of MIL-B-49430(ER).

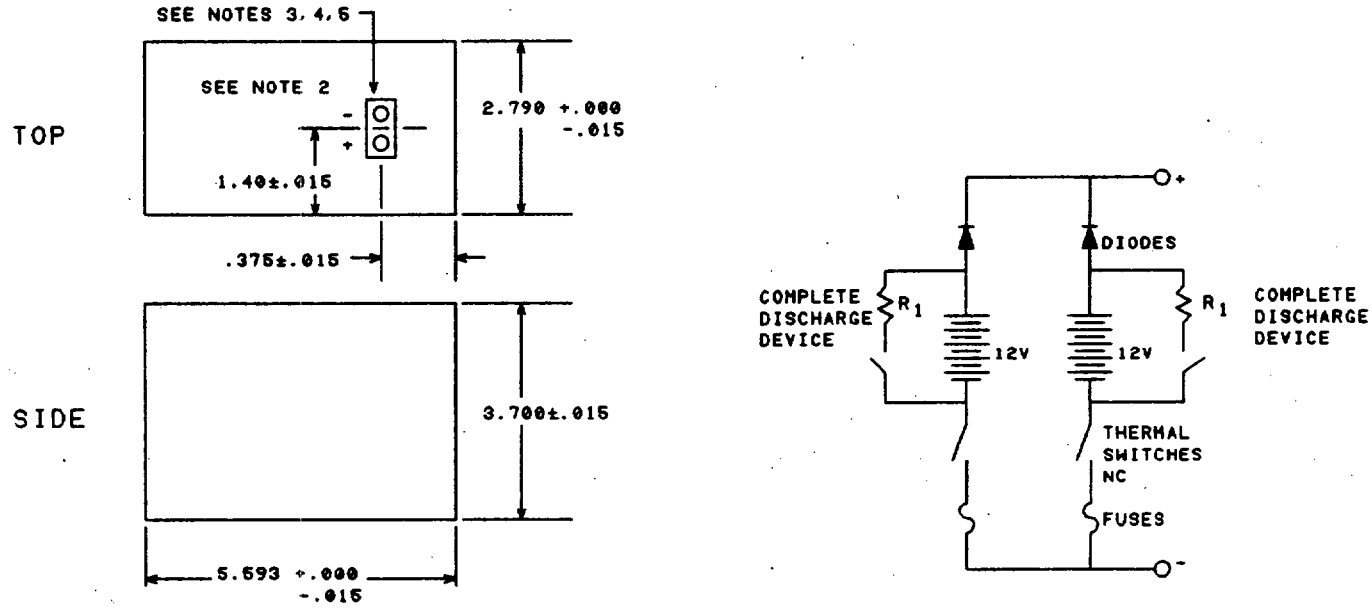


Figure 1. Battery dimensions and wiring schematic.

FSC 6135

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NOTES

1. All dimensions are in inches.

2. Connector wiring schematic to be printed or embossed on battery case in suitable location near the connector on the terminal face of the battery.

3. The connector socket shall be mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket when used initially, during and after subsequent insertions of the mating plug during contractual testing.

4. The battery terminal shall consist of a two hole female socket per MIL-B-18, Type I. The connector shall be mounted in accordance with referenced drawing.

5. Socket shall float 1/32 inch in all directions from the specified pin circle center location in the plane of the socket.

6. R_1 value shall be between 70 ohms and 100 ohms and have a power rating of 2.0 watts.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 15.25 volts.

Minimum voltage for capacity: 10.00 volts.

A 2.25 ampere, non-replaceable, slow blow fuse shall be used in the negative leg of each section (see schematic).

A diode shall be used in each leg (see schematic). See paragraph 3.5.5.2 of the Basic Specification.

Terminals: 2 hole female socket per MIL-B-18,Type 1.

Weight (maximum): 2.04 pounds (925 grams).

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Capacity test: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 10.00 volts
I	3.2
L	1.8
H	5.6
HT	5.4
LT	1.6
T	3.0
D	3.0
ILR	1000

Initial voltage delay: During the I, H, HT, T, D, and ILR discharge tests, initial voltages below the 10.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests initial voltage below 10.00 volts cannot exceed a 60.0 second duration.

High temperature Switch: A high temperature switch or thermal fuse shall be employed internal to the battery in each leg (see schematic), as described in paragraphs 3.5.5 and 3.5.5.1 of the Basic Specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in para. 3.5.5.3 of the basic specification.

METHOD OF EXAMINATION AND TESTS:

Capacity Tests:

(1) Storage - Details on storage conditions for all specified capacity tests are described in the basic specificaiton.(See 4.8.9).

(2) Discharge -

Cells: Discharge at 2.0 amperes constant current at $70\pm5^{o}F$ minimum of 3.00 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 1.0 ampere constant current for 6 hours into voltage reversal.

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Batteries: For all capacity tests except the H, HT, and ILR the batteries shall be discharged at 4.0 amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through 4.0 amperes constant current for 5 minutes followed by open circuit for 5 minutes. This cycle shall be repeated until the closed circuit voltage reaches zero volts. For the I capacity test the batteries shall be forced discharged at 2.0 ampere constant current for 6.9 hours immediately after zero volts is obtained. There shall be no venting, fire or explosion when batteries are force discharged. For the ILR test a fixed resistance of 1175 ohms shall be used to a cut-off voltage of 10.00 volts or 1000 hours.

Closed Circuit Voltage Test: Load resistance of 7.00 ohms shall be used. Voltage shall be above 10.00 volts within five seconds.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
Army-ER

Preparing Activity:
Army-ER

(Project 6135-A209-09)

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